

# CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

## Lab Solutions – Set 1

### Question 1

*Write a Pandas program to select the distinct department id from employees file.*

#### Python Code:

```
import pandas as pd

# Create sample employees dataset
employees_data = {
    'EMPLOYEE_ID': [100, 101, 102, 103, 104, 105, 106, 107],
    'FIRST_NAME': ['Steven', 'Neena', 'Lex', 'Alexander', 'Bruce', 'David', 'Valli',
                  'Diana'],
    'LAST_NAME': ['King', 'Kochhar', 'De Haan', 'Hunold', 'Ernst', 'Austin', 'Pataballa',
                 'Lorentz'],
    'DEPARTMENT_ID': [90, 90, 90, 60, 60, 60, 50, 60]
}

df = pd.DataFrame(employees_data)

# Select distinct department IDs
distinct_dept_id = df['DEPARTMENT_ID'].unique()

print("Distinct Department IDs:")
print(distinct_dept_id)
```

#### Output:

```
Distinct Department IDs:
[90 60 50]
```

## Question 2

Write a Pandas program to display the IDs of employees who have done two or more jobs.

### Python Code:

```
import pandas as pd

# Create sample job history dataset
job_history_data = {
    'EMPLOYEE_ID': [102, 101, 101, 200, 200, 176, 176, 200],
    'START_DATE': ['2001-01-13', '1997-09-21', '2001-10-28', '1995-09-17', '2001-03-24',
    '2006-03-24', '2007-01-01', '2002-07-01'],
    'END_DATE': ['2006-07-24', '2001-10-27', '2005-03-15', '2001-03-23', '2001-12-31',
    '2006-12-31', '2007-12-31', '2006-12-31'],
    'JOB_ID': ['IT_PROG', 'AC_ACCOUNT', 'AC_MGR', 'AD_ASST', 'AC_ACCOUNT', 'SA_REP',
    'SA_MAN', 'IT_PROG'],
    'DEPARTMENT_ID': [60, 110, 110, 90, 110, 80, 80, 60]
}

df = pd.DataFrame(job_history_data)

# Find employee IDs with 2 or more jobs
job_counts = df.groupby('EMPLOYEE_ID').size()
multi_job_employees = job_counts[job_counts >= 2].index.tolist()

print("IDs of employees who have done two or more jobs:")
print(multi_job_employees)
```

### Output:

```
IDs of employees who have done two or more jobs:
[101, 176, 200]
```